FOUNDATIONS OF AI & ML	
Course Code: CY51	**Credits: 2:1:0**
Prerequisite: Nil	**Contact Hours: 28L+14T**
Course Coordinator: Dr. Naveen N C	

Course Contents

Unit I

Introduction to AI & Agents: Why study AI?, What is AI?, The Turing Test, Rationality, Branches and Brief History of AI, Challenges for the Future, Intelligent Agents: Definition, Rational Agents, Rationality, Performance Measure, PEAS, Agent Types: Reflex, Model-based, Goal-based, Utility-based**.**

- Pedagogy: Chalk and board, Problem based learning, Visual Aids
- Link: https://nptel.ac.in/courses/106106126

Unit II

Search & Problem Solving: Problem-solving agents, problem formulation, Grid world and Vacuum world, Uninformed Search: DFS, BFS, DLS, IDS, Informed Search: Best-First, A*, Heuristic Search

- Pedagogy: Chalk and board, Problem based learning, Pair Programming
- Link: https://nptel.ac.in/courses/106106126

Unit III

Problem Reduction & Reasoning: Problem Reduction: AO* Algorithm, Game Playing: Minimax, Alpha-Beta Pruning, Introduction to Knowledge-Based Agents, Wumpus World, FOL

- Pedagogy: Chalk and board, Problem based learning, Role Play
- Link: https://nptel.ac.in/courses/106106126

Unit IV

Fundamentals of Machine Learning: Supervised vs. Unsupervised Learning, Evaluation Metrics, Hypothesis Space, Inductive Bias, Linear Regression, Decision Trees, Overfitting, KNN, Cross-validation

- Pedagogy: Chalk and board, Problem based learning, Case Studies
- Link : https://nptel.ac.in/courses/106106202

Unit V

Advanced ML Techniques: Logistic Regression, SVM, Kernel SVM, Neural Network: Perceptron, MLP, Backpropagation, Introduction to DNN, Clustering: K-Means, Hierarchical Clustering

- Pedagogy: Chalk and board, Problem based learning, Case Studies
- Link: https://nptel.ac.in/courses/106106213

Text Books:

1. Artificial Intelligence-A Modern Approach, Stuart J. Russell and Peter Norvig, Pearson 4th Edition, Eleventh Impression 2022.
2. Tom Mitchell, Machine Learning, First Edition, McGraw- Hill, 1997

Reference Books:

1. Ethem Alpaydin, Introduction to Machine Learning, 2nd Edition, MIT Press.2010.
2. Elaine Rich, Kevin Knight, Shivashanka B Nair: Artificial Intelligence, Tata MCGraw Hill 3rd edition. 2013
3. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, 2nd Edition, Aurelien Geron, O'Reilly

Course Outcomes (COs):

At the end of the course student will be able to:

1. Understand foundational concepts, history, and scope of Artificial Intelligence. (PO-1,2,3 PSO-1)
2. Analyze AI solutions using search and agent models. (PO-1,2,3,4,5 PSO-1,2)
3. Demonstrate logical reasoning and game-playing using suitable algorithms. (PO-1,2,3,4,5 PSO-1,2)
4. Apply machine learning algorithms to classification and prediction tasks. (PO- 1,2,4,5 PSO-1,2)
5. Evaluate advanced models using neural networks and clustering techniques. (PO-1,2,3,4,5 PSO-1,2)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment Tools	**Marks**	**Course Outcomes addressed**
Internal Test-I (CIE-I)	30	CO1, CO2
Internal Test-II CIE-II)	30	CO3, CO4, CO5
The average of the two CIE shall be taken for 30 marks		
Other Components		
Tutorial Assignment-1	10	CO1, CO2, CO3
Tutorial Assignment-2	10	CO3, CO4, CO5
The Final CIE out of 50 Marks = Average of two CIE tests for 30		
Semester End Examination (SEE)	100	CO1, CO2, CO3, CO4, CO5